# Performance Sensitivities of Wireless Mesh Networks Under Path-based DoS Attacks

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***Abstract*—This paper examines the performance of wireless mesh networks (WMNs) under the impact of path-based denial of service (DoS) attacks. Specifically, we study the factors that are conducive to path-based DoS attacks, while focusing external interferences, medium errors, and physical diversity. We setup a wireless mesh testbed and configure a set of experiments to gather measurements and assess the effects of different factors. We find that the impact of external interferences and medium errors on network performance is exacerbated when path-based DoS attacks are carried out. Another interesting observation is that a far attacker can lead to an increased performance degradation than a close-by attacker due to physical diversity. Further, we discuss a simple strategy to counter path-based DoS attacks, which has potential for reducing the impact of the attack significantly.**

## I. Introduction

Wireless mesh networks (WMNs) are emerging as next generation wireless networks that allow inexpensive 802.11 radios to be used for extending wireless coverage area and for offering cost effective and highly reliable communication. However, the easy access and multi-hop communication nature of such networks surrender themselves to a wide variety of attacks, either passive (such as eavesdropping or traffic analysis) or active (such as message modification, replay or denial of service (DoS)). Moreover, the distinguished architecture of WMNs [2], e.g., using mesh routers and mesh clients, may have a significant impact on the responses of WMNs to these attacks, in compared with wireless LANs and mobile ad hoc networks (MANETs). In this paper, we study the performance impact of DoS attacks in that their primary goal is to cripple the performance of networks. This study will help us understand the level of performance, WMNs can offer to applications under path-based DoS attacks. In addition, it will also assist us to deduce the impact of the prevalent factors in WMNs, such as external interferences, medium errors and physical diversity, and how these factors can be conducive in increasing the impact of path-based DoS attacks.

Path-based DoS attacks have been studied in the context of sensor networks in [6]. However, the WMNs are very different than sensor networks in a way that an attacker has to attack much more powerful nodes in WMNs as compared to low processing and low memory nodes in sensor networks. As the multi-hop flows are inherent in WMNs, it becomes very important to view the path level impact of such attacks. We believe that our work is useful for the design of attack detection algorithms, and also for providing corrective measures for the efficient placement of routers in WMNs.

*This project is supported by National Science Foundation (NSF) under award ECS-0524519.

In particular, we focus on WMN performance under the attacks by changing wireless cards power levels (medium errors), by considering attacker's mobility (physical diversity) and in the presence of external wireless devices (interference), which cannot easily be investigated with simulations and analysis. We find that i) the presence of *hidden* poor links (due to retransmission at MAC layer) in a network is beneficial to path-based DoS attacks, though weak radio signals are not favorable to both legitimate users and attackers; ii) the presence of interference induces much higher packet losses even due to a low intensity path-based DoS attack; iii) an attacker farther from the targeted path can be more damaging than the attacker in the close proximity of the targeted path because of physical diversity; iv) a simple countermeasure strategy has potential for reducing the impact of the attack significantly.

The remaining paper is organized as follows. In Section II, we discuss existing works about possible attacks on WMNs. The goals of our study, its challenges, characteristics of attacks, and testbed setup are discussed in Section III. In Section IV, we describe each factor, and present the results of our measurement study with a summary in the last. A discussion on the design of a simple countermeasure is presented in Section V. Finally, Section VI concludes the paper.

## II. Background and Related Work

Attacks in wireless networks can be broadly categorized as passive and active. Eavesdropping and traffic Analysis are passive attacks, in which an attacker overhears an on-going communication [3]. As the goal of passive attacks, in general, is to steal the confidential information without impacting the performance, we do not focus them in our work. Message modification, replay attack, and denial of services (DoS) are kind of active attacks which targets the performance degradation of on-going communications (or flows) in addition to other malicious intentions. Message modification and replay attacks achieve their goal by targeting weaknesses in a security protocol, and may be thwarted by using stronger security protocols. In contrast, DoS attacks reduce the bandwidth used by the legitimate nodes, and are not possible to curb even by strong cryptographic mechanisms. As DoS attacks severely impact the performance and can be carried out rather easily compared to other attacks, we address them in our paper.

The impact of DoS attacks on wireless networks has been studied extensively because they can be carried out in various ways, such as by exploiting the vulnerabilities in IEEE 802.11 wireless protocol [4]; by following non-compliant protocol behavior [8]; or by creating high RF noise such that neighboring nodes are not able to access the channel (physical

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layer jamming) [12], [10], [9]. For instance, Bellardo et al. [4] studied the impact of deauthentication and virtual carrier sense[1] attacks, and also discussed simple and elegant solutions to thwart them. Similarly, the JellyFish and black hole DoS attacks at network layer are studied in [1]. It is noticed that attackers in these studies are protocol compliant, but carry out attacks in a way which leads to DoS to the legitimate users. Raya et al. [8] present IEEE 802.11 protocol misbehavior in which an attacker modifies protocol parameters (such as *NAV* value) to gain high bandwidth share. Extensive research effort in the past is spent to understand, detect, and counteract such attacks in wireless networks [7], thus it is not of our interest in this paper. Inherently, DoS attacks due to high RF noise create high interference such that the neighboring nodes are denied access to the channel. Xu et al studied four different jamming models and their effectiveness in wireless networks, although not providing any solutions to prevent them [11]. Similarly, jamming of encrypted wireless ad hoc networks at transport (TCP) and network (AODV) layers is studied in [5].

However, the performance impact of DoS attacks on real-time traffic in *real* world remains an open question, especially for mesh networks. Therefore, in our study, DoS attacks are generated as traffic load in a way that prevents legitimate flows from using their full capacity leading to degradation in the network performance. Specifically, we focus on MAC layer DoS attacks due to following reasons: 1) these attacks are very easy to be generated either by internal or external nodes in a network; 2) the attackers are protocol complaint while generating attack traffic; 3) such DoS attacks are difficult to be thwarted by the victim node. The very architecture of the IP layer is such that any node can originate traffic toward any destination and the destination node is helpless in stopping the traffic before it reaches the network. The fundamental characteristics of WMNs also facilitate an easy ground for conducting such DoS attacks. For protocol complaint DoS attacks, each attacker targets more than one nodes on the path (therefore named as *path-based DoS attacks*) to inflict sever damage to an on-going flow. We investigate the impact of these attacks on the performance of WMNs in different scenarios in this work.

## III. Goals and Methodology

The goal of this study is to assess the performance of WMNs and further understand how well they work under path-based DoS attacks in realistic settings. In particular, we are interested in the results of interaction among various factors, such as external interferences, medium errors and physical diversity. To this end, we perform a small-scale measurement study in order to answer following questions:

- *How does the physical diversity affect the impact of path-based DoS attacks?* Wireless links experience varying medium conditions depending upon their locations. It is not yet discussed in the literature that how these varying conditions can interact with the path-based DoS attacks.
- *Are the medium errors conducive to path-based attacks?* Medium errors are generally very common on wireless links. The retransmissions at the MAC layer help in reducing the impact of medium errors. However, we are interested in understanding about how the performance is impacted by the path-based DoS attacks, when varying medium errors are present on the links.
- *What is the impact of path-based DoS attacks when external interferences are present?* It is obvious that the performance of WMNs degrades in the presence of path-based DoS attacks. However, it is not clear that what the performance trend is under external interferences.

[1]For brevity, we denote carrier sense as CS in this paper.

### A. Challenges

Our experiments are designed to answer the questions posed earlier. We use different network configurations and settings to tease apart the effects of the discussed factors, and fine-grained measurements to assess the WMN performance under the path-based DoS attacks. Throughout this study, various factors, such as interference, medium errors, and physical diversity, help us in simulating different scenarios experienced in real environments. All factors are helpful in providing deeper insights into the obtained experimental measurements.

One of the main challenges in our study is placing routers in locations so that an attacker router can be outside CS range of particular routers whenever needed. Due to geographic restriction in placing routers anywhere in the campus building, we fix the router positions, which is typical the design of WMNs [2]. However, the network connection are changed by varying power levels of these routers. Another challenge is that we aim to differentiate the effects of routing and attacks, e.g., packet losses due to the attack or link dynamics. This is not trivial because we observe that paths are updated frequently, even in our quite limited-size setup with OLSR protocol. It becomes very difficult to have consistent observations unless we can have stable paths. Therefore, we resort to static routing for achieving stable paths in order to provide accurate observations.

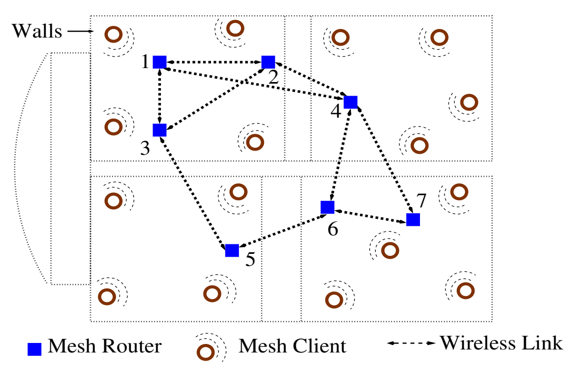


Fig. 1. Testbed Setup.

### B. Characteristics of Attacks

To obtain the maximal effect of an attack, the location of the attacker node is carefully chosen such that it is outside the CS range, but inside interference range of all of the routers en-route a path. Therefore, the attacker acts as a hidden terminal to the sender and can inflict the significant damage to the sender traffic. Moreover, as it is outside CS range of all intermediate


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routers and receiver node too, the attacker node can also interfere with the traffic forwarded by intermediate nodes in the similar manner, and same with the traffic coming back from the receiver. Therefore, the attacker is able to cover the maximum portion of the path between sender and receiver to maximize the impact of attack. It is to note that the attacker is protocol compliant, and follows all rules at the MAC layer.

### C. Wireless Mesh Testbed

We setup a wireless mesh testbed as a miniature version of infrastructure wireless mesh networks deployed in real environments. The testbed is on the second and third floor in a building for which a sketch diagram is shown in Fig. 1. The testbed consists of 7 mesh routers, and several mesh clients to form a wireless multi-hop mesh network. The 3 mesh routers (with ids 2, 5, 7) also act as access points. The clients are mobile, whereas all mesh routers are static in our testbed. All nodes (routers or clients) are using 802.11b based wireless cards. The routers are configured in *ad hoc* mode, and using a *predetermined* channel, which is the farthest than the channels used by other networks in the vicinity of our testbed to reduce the impact of interference from those networks. The transmission rate on all nodes is set to 11 Mbps. The *ip* addresses are assigned statically to each node to eliminate the delay due to DHCP configuration. All nodes are running Ubuntu linux with 2.6 kernel. In addition, one machine is used as controller and helps in accessing all nodes (mesh routers and clients) remotely over our campus wired network for monitoring and configuring multiple measurements during a long period time.

The details of traffic flows are as follows, which are the same for all experiments, otherwise specified for each individual scenario throughout the paper:

**Attacker**: Router 1 is selected as the attacker node which broadcasts traffic in the network. The power levels of routers are set in such a way that it is out of CS range of and inside interference range of the routers on the flow, and can act as attacker for the flow.

**Starting and ending routers**: We randomly take a pair of client nodes for measurements, but focusing on a particular pair of router nodes indexed 5 and 7 for discussion only, where router 5 is the end point or the egress router and router 7 is the starting point or ingress router in the network, although the clients associated with routers 7 and 5 are *actual* sender and receiver. We compute an average of experimental measurements by considering multiple client pairs for better accuracy.

**Routing and locations**: Router 6 acts as the intermediate node to enable a minimum of 2-hop. The location of each router is chosen to ensure that packet losses between a client and an access or ingress router are almost negligible; thus measuring packet losses occurred only in the backbone (formed by routers) of wireless mesh network.

**Traffic flows**: The flow between routers 5 and 7 is ICMP traffic, whereas attacker broadcasts UDP traffic. The attacker's packet rate and packet size are kept fixed at 100 packets/sec (pps) and 200 bytes IP packet[2] in this scenario.

[2]Just for differentiation, the attacker's packet size is the same as IP packets, whereas the normal flow packet size is the only the application payload.

## IV. Measurements and Observations

The main objective of this paper is to present our observations of the WMNs performance in response to path-based DoS attacks. To demonstrate the strengths of this experimental work, we focus on several factors that have not been studied in simulations and analysis due to the device and environmental complexity at hand.

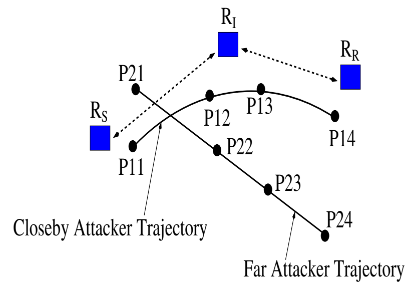


Fig. 2. Attacker's Mobility.

### A. Physical Diversity (Attacker's Mobility):

Physical diversity can arise in wireless networks because of dynamic radio channels in real world. For example, when wireless devices are at different locations, the physical channel associated with devices can become affected differently by the adjacent environment, even though devices are using same physical channel. Also, multiple devices (or users) interact with each other for obtaining shared medium, leading to different channel conditions. In this work, we focus on the effects of *the same channel*, without considering the effects of channel assignment.

We study physical diversity by moving an attacker in different directions (by keeping other parameters, such as packet rate and size constant) to measure the variations in their impacts. We do so because, if the attacker's position is fixed, the impact of the attack may also be fixed if we assume that the other conditions remain similar during the execution of the attack. However, an intelligent attacker would want to move in such a manner nearby the targeted path to inflict the most severe damage. Although, it may seem obvious that if the attacker moves very close to the targeted path, then the impact of attack may be higher. However, we find that it may not be always true and when the attacker moves farther from the targeted path, the impact of the attack may be higher.

We use two experiments for this study. In the first experiment, the attacker (closeby attacker in Fig. 2) moves along the targeted path, and we measure the packet losses at four different locations, where P11 and P14 are starting and end points, and P12 and P13 are any intermediate points. In the second experiment, the attacker (far attacker in Fig. 2) moves away from the path, and packet losses are measured at four locations, where P21 and P24 are starting and end points, and P22 and P23 are any intermediate points in Fig. 2.

**Observations:** The results are presented in the Fig. 3. We notice that *the packet losses are higher when the attacker moves away than the packet losses when the attacker moves closer to the targeted path.* The reason is that when the (closeby) attacker

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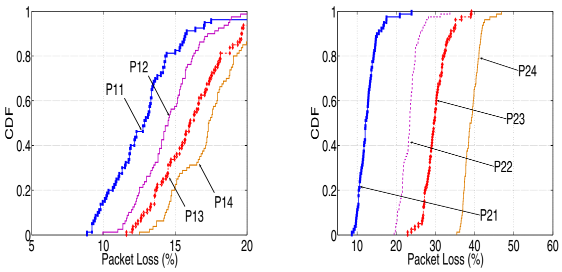



(a) Close-by Attacker.

(b) Far Attacker.

Fig. 3. Impact of Attacker's Mobility.

is very close to the targeted path, the attacker remains in the CS range of several nodes on the path. Therefore, the impact is not very severe except stealing some of the capacity of the path. Consequently the impact from all the four points are similar. However, when the attacker (far attacker case) moves away from the path, initially (at the first point P21) the impact is very similar to the impact when the attacker moves closely. It is because the attacker is still in CS range of many nodes on the path. However, as the attacker moves further away, the impact rises and the packet losses increase. We notice that the packet losses are highest at the point P24. It is because that the attacker is completely out of CS range of any of the nodes on the path, therefore causes the highest damage. Although, the packet losses are high at P22 and P23 points too, we observed (by running a separate small experiment) that the attacker was not completely out of CS range of all the nodes on the path. Infact, the attacker was able to sense the channel busy intermittently at these points. Due to this, the impact of attack is less as compared to the impact when the attacker is at the point P24.

The important conclusion is that it is not always good to create path which are far from the attacker's position. One way is to form paths such that the attacker is outside the interference range of all the nodes so that the attacker does not act as an attacker at all. However, if it can not be possible always, then one of the strategies to counter it can be to form the required path as close as possible to the attacker. In this way, the path will lose some of its capacity (due to carrier sensing with the attacker) but packet losses will be low and the impact of the attack will be very minimal.

### B. Medium (Radio Link) Errors

Wireless devices employ error correction mechanisms at the physical layer to minimize the effect of medium errors. For example, if the transmission is not successfully received, a negative acknowledgment (NACK) is sent to the sender by the receiver. Once the sender receives a NACK, it retransmits the frame. Often time due to retransmissions at MAC layer, the actual quality of poor links is not exposed. We call such links as *hidden poor links*. Therefore, it becomes essential to study that how a path-based attack behaves when such poor links are present in a network. To study this, we vary power levels of the nodes in our testbed to generate different radio link errors. Specifically, we used power level values of 14, 13, 12, and 11 dBm. Initially, we set the retry count to 0 to obtain actual errors on the path, and there was no attack carried out. The measurements are presented in the Fig. 4.

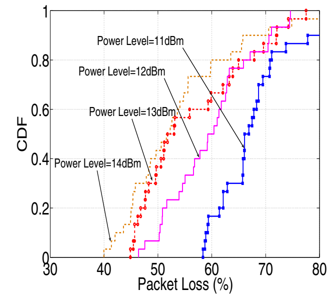


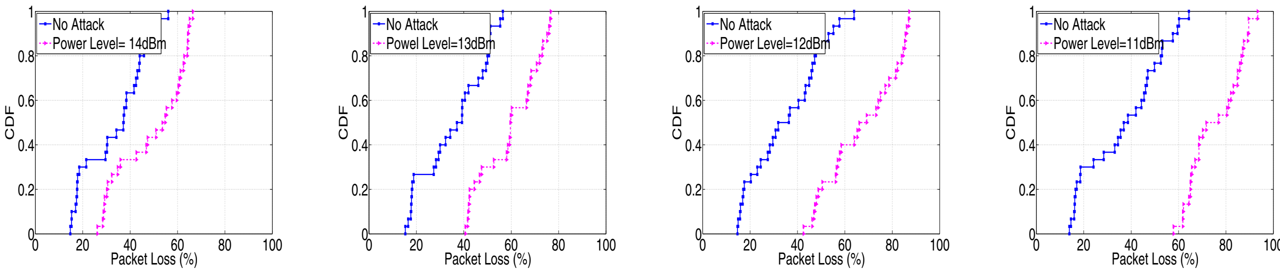

Fig. 4. Packet Loss CDF: No Attack (retry=0).

We notice that as the power level is reduced, there are higher packet losses experienced on the path. It shows that reduced power level led to higher radio errors on the links and caused higher packet losses. Now, we carried out the same experiment again but with retry count set to the default value (7 in our wireless cards). We noticed that packet losses under the above power level values were almost similar, and were close to 20%. It is because that the retranmissions masked the impact of poor quality of the path induced due to reduced power levels. However, we will show that as soon as the same experiment is carried out again under the presence of attack, the attack takes benefit of *hidden radio errors*, and causes higher packet losses with low power levels as compared to high power levels. Experimental setup is similar to the previous scenario, except that power levels are varied to the above specified values when attack is carried out, and retry count is again set to the default value. The results are presented in Fig. 5.

It can easily be seen that the path experiences similar packet losses without attack even though power levels of the links are varied (i.e., due to retransmissions). However, when the attack is carried out, there are different packet losses experienced with different power values. The packet losses increase monotonically as the power level is reduced from 14 to 10. It is evident from the results, that the radio errors which were *masked* due to retransmissions, are exposed again, and they are conducive to the attack. Therefore, our results suggest that *path-based DoS attack is more effective on the path with* hidden *poor links.* An attacker (either external or internal) can easily take advantage of this fact, and might make the network disconnected, if the attacker acquires the network situation. This observation reveals that it is essential to protect *weak paths* as compared to the paths with very good quality links.

### C. External Interference

Interference is pervasive in wireless networks as signals from different sources impact each other. It is even of more concern in WMNs as the traffic from different sources can affect routing leading to degradation in the quality of multi-hop paths. In WMNs, interference can be caused by internal sources, e.g., routers inside the network, or external sources, e.g., other networks or electronic devices such as mobile phones, microwave oven etc. The impact of interferences in




(a) Power Level=14dBm. (b) Power Level=13dBm. (c) Power Level=12dBm. (d) Power Level=11dBm.

Fig. 5. Power Level Variations.

WMNs due to internal sources has been studied extensively in the past [11]. However, it is still not very clear how a path-based DoS attack can take advantage of external interferences.

To understand the interaction between external interference and path-based DoS attacks, we ran experiments during nights, for minimal interference. It is almost impossible to do experiments without any interference, even if there are very few people present. Therefore, our measurements have minimal impact of interference which we consider as the *base measurements*. Moreover, we also ran some of our experiments during day time to explicitly study the impact of external interference and to compare them with the measurements taken over nights.

We present results showing packet losses with *no-attack* and *attack* under no interference in Figs. 6(a) and 6(b), respectively. We observe that packet losses under no attack are below 15% and are almost similar across different packet sizes. However, they are mostly upto 35% under attack. Also, we notice that the attack is not very effective for small packets, but it is more severe in cases of large packets such as 256 or 512 bytes, causing almost 3 times more packet losses as compared to the packet losses under no attack.

**Observations:** Fig. 6(c) shows the impact of attack when there are external interferences in the surrounding environment. These measurements are collected at the day time with more humans around, and there are significant external interferences due to heavily used campus wireless LANs, electronic devices in the proximity, etc. We notice that impact of attack is more than twice in the presence of interferences. It is because to handle the volume of unsuccessful packets as soon as an attack is carried out, the packet buffer is increased rapidly as many packets are not successfully transmitted. Therefore, MAC layer buffer, which is designed to store failed packets for retransmissions, reaches its capacity faster, and the packets can be dropped. We note that this phenomenon is also possible without interference if the attack is carried out at very high intensity. However, in the presence of interferences, a mild intensity (like 100 pps attack in this experiment) attack can also cause very high packet losses.

### D. Summary

In this work, we have focused on conducive nature of the prevalent factors (such as physical diversity, radio link errors, and external interferences) toward path-based DoS attacks. Now, we present a summary of our observations, and discuss their implications to real world scenarios.

**Remark 1:** We have shown that physical diversity may help attackers to cause severe damage even if the attackers are relatively far from the targeted paths. Our observations suggests placing routers in such a manner that each router is either outside interference range or inside CS range of other routers in the network.

**Remark 2:** We observed that retransmissions at the MAC layer can mask the *hidden* poor links in a network. However, when the attack is carried out, these poor links are exposed and contribute toward very high packet losses. Although, retransmissions at the MAC layer are necessary to reduce the packet losses, the fixed retransmission value may not be appropriate. Based on our observations, it is necessary to adjust the retransmission to a higher value to protect those poor links from getting exposed during an attack. Of course, increasing the value of retransmissions may not be good for delay sensitive applications. We believe that it is essential in the future to adjust the retransmission value dynamically for optimal performance of WMNs.

**Remark 3:** We showed that interferences can increase the effect of a low intensity attack. We found that the observation is due to the fact that there is a limited size buffer at MAC layer to hold packets for retransmissions. Therefore, it is essential to have a dynamic buffer size to reduce the impact of attack in the presence of the interferences. A larger buffer can help in reducing the packet losses. However, an optimized size of the buffer depending upon the intensity of the attack and interference should be more appropriate than just choosing arbitrary large value because of its negative impact on packet delay. We do not study on finding an optimized size of the buffer in this work, and leave it for future work.

## V. Design of a Countermeasure

We have, by far, discussed the impact of attack by changing their intensity on the normal flows in WMNs. Now we discuss a simple strategy to lessen the impact of attacks studied above. In this strategy, a *helper node* is placed outside the interference range of routers en route normal flow path, and inside the interference range of attacker node. In this way, the helper node acts as an attacker for the attacker node, but does not impact the normal flows.

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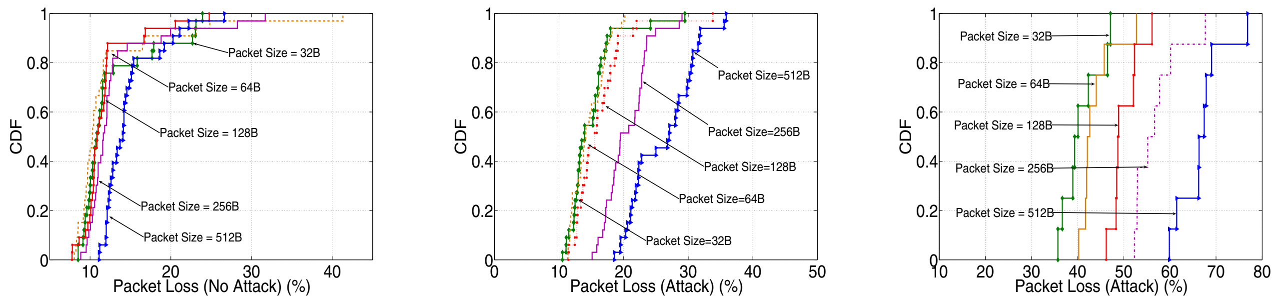


(a) No Attack.

(b) Attack (No Interference).

(c) Attack (Interference).

Fig. 6. Packet Loss CDF. Effects of Packet Size.

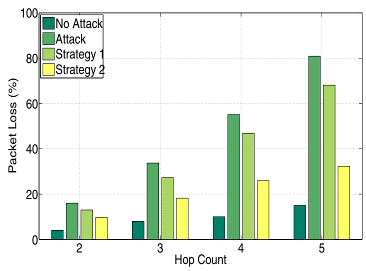


Fig. 7. Countermeasure Strategy: Helper Node.

Specifically, we experimented with 2-, 3-, 4- and 5-hop paths in our testbed. The normal flow on the path consists of 64 bytes packet size and packet rate of 100 pps. The packet size and rate of the attack traffic is set to 200 bytes IP packet, and 500pps, respectively. In addition to that, a helper node is placed in two ways in the testbed. In the first case (strategy 1 in Fig. 7), the helper node is placed inside CS range of the attacker node. Whereas in the second case (strategy 2 in Fig. 7), the helper node is placed outside CS range and within interference range of the attacker node. The traffic parameters of the helper node are similar to the attacker node. We observe from results in Fig. 7 that both strategies reduce the damage caused by the attack. However, strategy 2 is much more powerful than strategy 1, as traffic from the helper node collides with the attack traffic significantly leading to reduction in packet losses for the normal flow by more than half of the packet losses caused by the attack. Therefore, it shows that such counter strategies can be useful in increasing the performance against path-based DoS attacks. This observation suggests a helper node may be elected when a path-based DoS attack is detected.

## VI. Conclusions

We have presented a measurement study on the impact of path-based DoS attack in WMNs. Through the testbed setup and traffic generation tools, we have shown how various factors, such as physical diversity, radio link errors, and external interferences, affect the impact of path-based DoS attacks on the performance of WMNs. Our results suggest that a far attacker can be more dangerous than a close attacker due to physical diversity. Due to the retransmissions at the MAC layer, the hidden poor links become exposed when a path-based DoS attack is carried out. Both of these factors are very conducive to the path-based DoS attacks, and lead to significant packet losses. In addition, external interference also helps in exacerbating the impact of low density attack. We also discussed a strategy, where a node can act as a countermeasure against an attacker, and can help reduce its impact, if the location of the attacker is known. As a future goal, we plan to extend our study to a larger testbed to obtain more generic impact of attack traffic, while this study acts as a first step toward that goal.

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